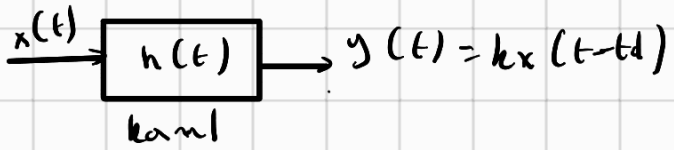
Bozulmasız iletim

(Distortionless Transmission)



$y(t) = K \cdot x(t-t_d)$

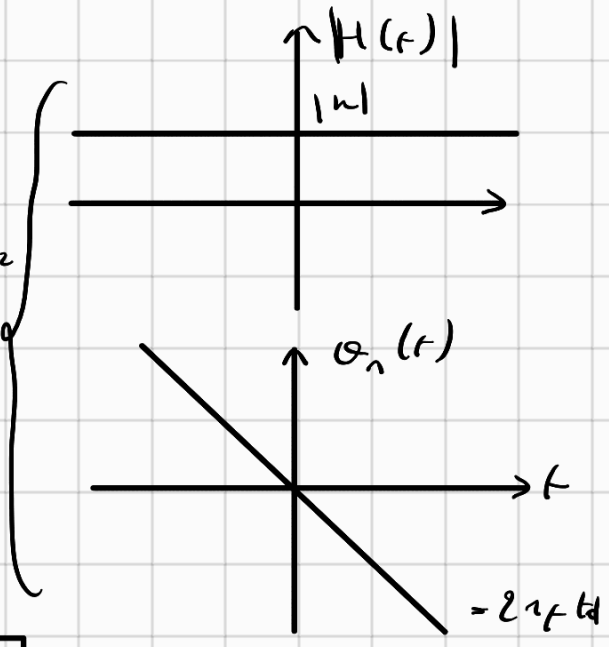
$Y(f) = k\,X(f)\,e^{-j2\pi f t_d}$

$H(f) = \frac{Y(f)}{X(f)} = k\,e^{-j2\pi f t_d}$

$= |H(f)|\,e^{j\theta_h(f)}$

$\Rightarrow \begin{cases} |H(f)| = |k| \checkmark \\ \theta_h(f) = -2\pi f t_d \checkmark \end{cases}$

Bozulmasız sistem



F^{-1}

$\star\ h(t) = k\,\delta(t-t_d)$

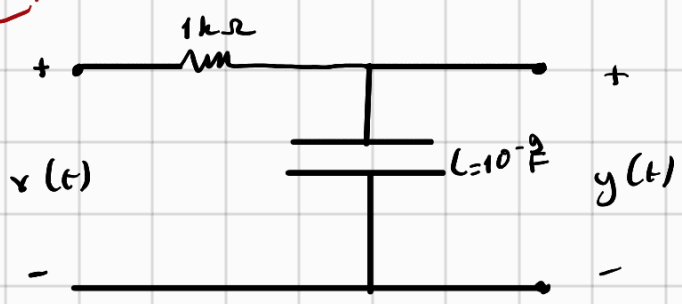
Grup gecikmesi $= t_d(f)$

$= -\frac{1}{2\pi}\frac{d\theta_h(f)}{df}$

(Group/envelope delay)

$\theta_h(f) = 2\pi f t_d \Rightarrow t_d(f) = t_d$ ↓ sabit ★

Örnek:



Bu filtreden işaretin bozulmadan geçmesi için x(t)'nin bant genişliği ne olmalıdır?

(%2'lik genlik cevabı toleransı, %5'lik grup gecikmesi toleransı mevcuttur.)

Alçakgeçiren filtre

$$H(f) = \frac{Y(f)}{X(f)} = \frac{\frac{1}{j2\pi fC}}{R + \frac{1}{j2\pi fC}} = \frac{1}{1+j2\pi fRC} = \frac{1}{1+j\frac{f}{a}}$$

$$= \frac{a}{a+jf}$$

$$\boxed{a = \frac{1}{2\pi RC} = \frac{10^6}{2\pi}}$$

$$\rightarrow |H(f)| = \frac{a}{\sqrt{a^2+f^2}}$$

$$\theta_h(f) = 0 - \arctan\left(\frac{f}{a}\right)$$

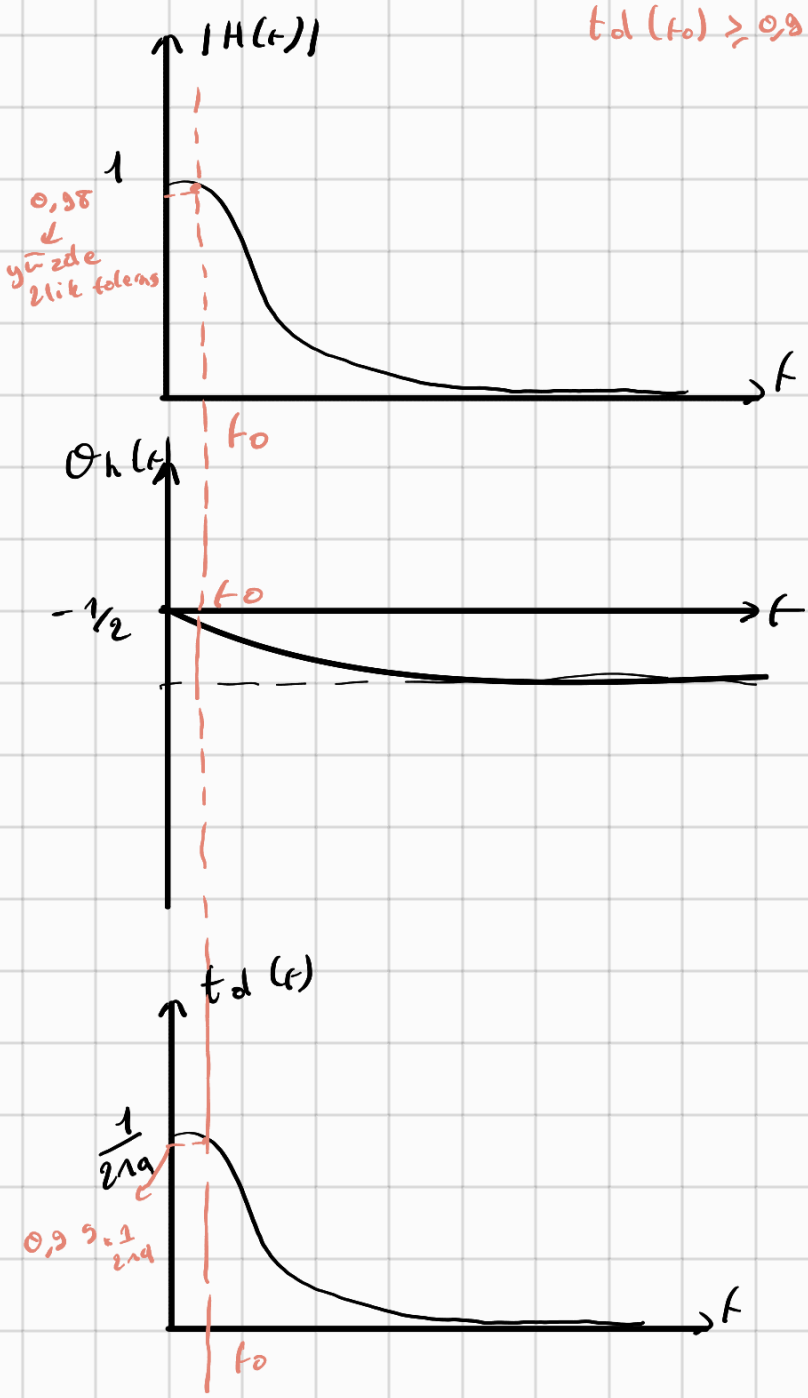
$$t_d(f) = -\frac{1}{2\pi}\frac{d\theta_h(f)}{df} = \frac{1}{2\pi}\frac{a}{a^2+f^2}$$

$z = \frac{y}{x} \rightarrow |y|e^{j\theta_y}$; $|x|e^{j\theta_x}$; $|z|e^{j\theta_z}$

$|z| = \frac{|y|}{|x|}$ $\quad \theta_z = \theta_y - \theta_x$

$|H(0)| = 1 \rightarrow |H(f_0)| \geq 0{,}98$ $\quad t_d(0) = \frac{1}{2\pi a}$

$t_d(f_0) \geq 0{,}95 \cdot \frac{1}{2\pi a}$



$|H(f_0)| \geq 0{,}98$

$\frac{a}{\sqrt{a^2+f_0^2}} \geq 0{,}98 \Rightarrow f_0 \leq 32{,}32\,kHz$

$t_d(f_0) \geq 0{,}95 \times \frac{1}{2\pi a}$

$\frac{1}{2\pi}\frac{a}{a^2+f_0^2} \geq \frac{0{,}95}{2\pi a} \Rightarrow f_0 \leq 36{,}5\,kHz$

$\Rightarrow f_0 = 32{,}32\,kHz$

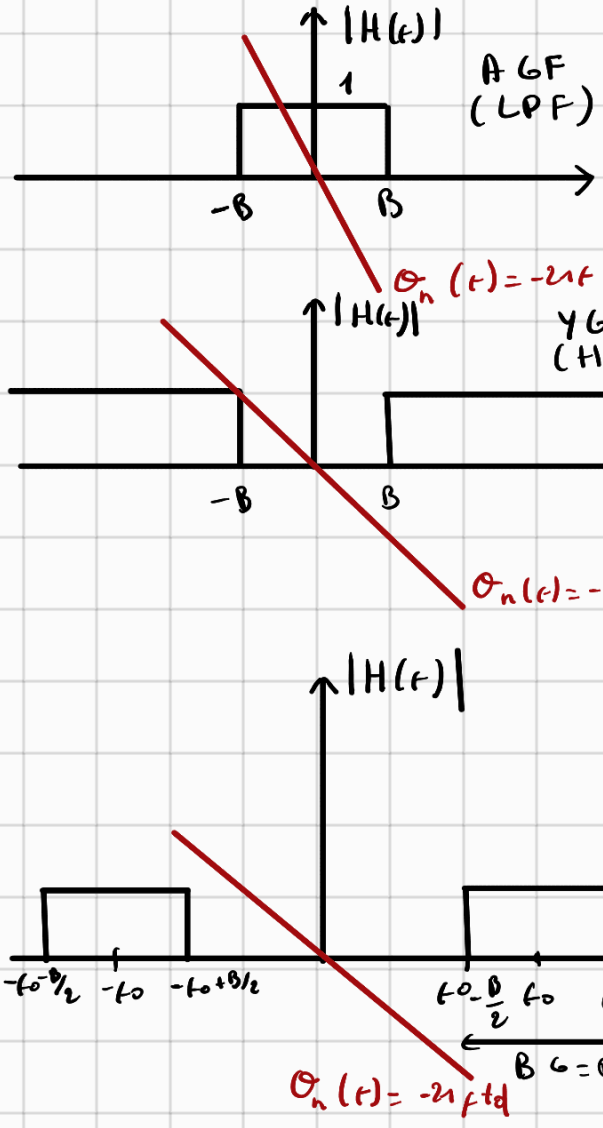
Bu bandda $(f < f_0)$,

$|H(f)| \cong 1$ $\quad t_d \cong \frac{1}{2\pi a} = 10^{-6}\,sn$

$y(t) = k \cdot x(t-t_d)$

$y(t) \cong x(t-t_d) = x(t-10^{-6})$

İdeal ve pratik filtreler



$BG\{x(t)\} \le B \Rightarrow y(t) = x(t-t_d)$

$|H(f)| = \sqcap\left(\frac{f}{2B}\right)$

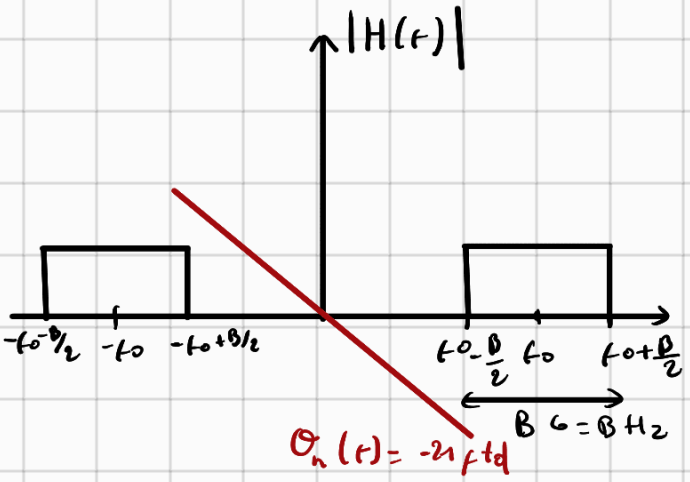
$H(f) = |H(f)| \cdot e^{-j\Theta_n(f)}$

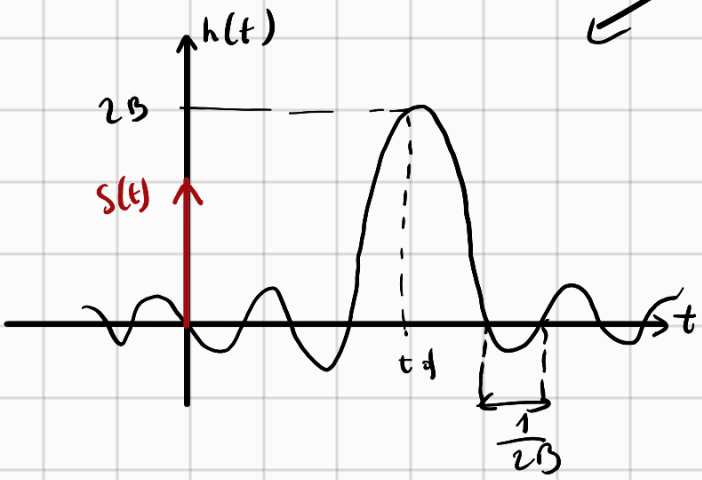
$H(f) = \sqcap\left(\frac{f}{2B}\right) e^{-j2\pi f t_d}$

$\downarrow F^{-1}$

$h(t) = 2B\,\mathrm{sinc}[2B(t-t_d)]$

$t = \frac{n}{2B} \quad 2Bt = n \in \mathbb{Z} - \{0\}$





Bu sistem nedensel değildir çünkü $h(t) = 0,\ t < 0$ şartı sağlanmıyor!!

çok büyükse

$t_d \gg$ $h(t)$ nedenseldir

$t_d \approx 3 \cdot \frac{1}{2B}$ ise $h(t)$ ideale yakın ve nedensel bir gerçekleştirmedir.

Ses işareti : $B = 20kHz \Rightarrow t_d \simeq 3 \frac{1}{2 \cdot 20 \cdot 10^3} = 0,75 \cdot 10^{-4} = 7,5 \cdot 10^{-5} \approx 10^{-4} = 0,1\,ms$

- $\hat{h}(t) = h(t) \cdot w(t)$

Maliyeti Bant genişliği arttırır

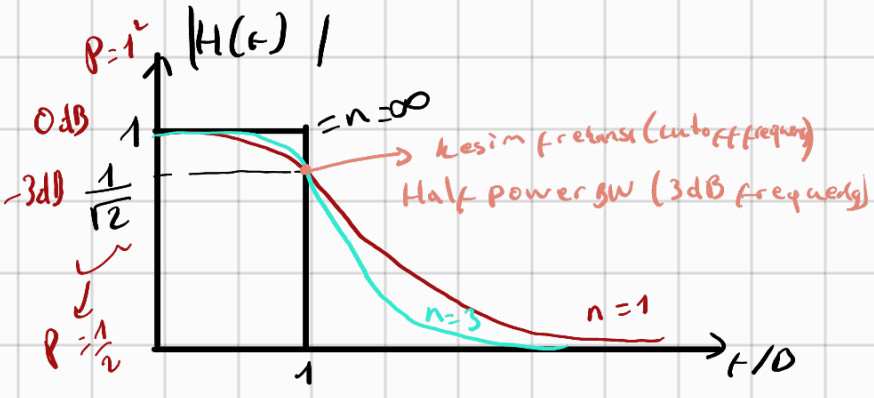
0'daki keskin geçişinden dolayı yüksek frekanslarda bileşenleri var

diğer çözüm $w(t)$ pencereleme ile çarparız

$\hat{h}(t) = h(t) \cdot w(t)$

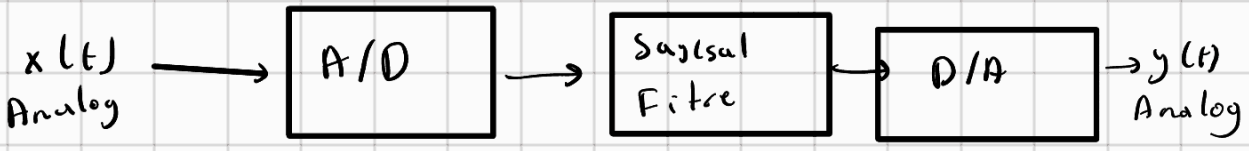
Pratik filtre: Butterworth

$$H(f) = \frac{1}{\sqrt{1 + (f/B)^{2n}}}$$



n: Filtrenin derecesi

n arttıkça filtrenin karmaşıklığı artar ve gecikme artar hemde maliyeti artıyor



Bir haberleşme kanalında işaret Bozulması

- Lineer Bozulma
(Linear Distortion)

kanal ideal ⇒ Bozulmasız (Distortionless)

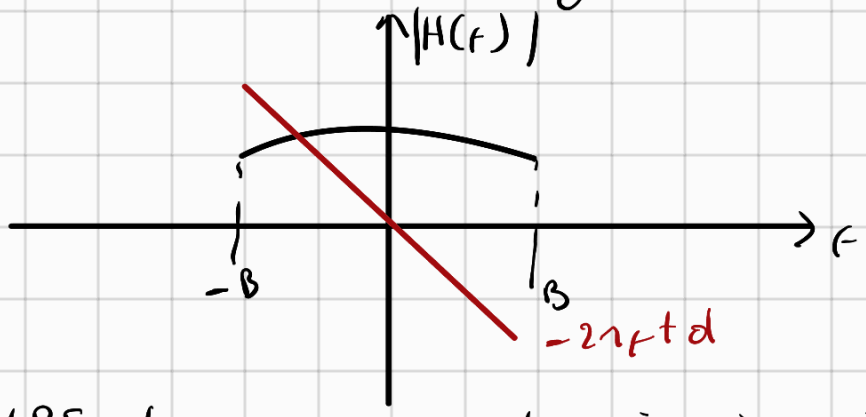
$|H(f)| = |k|$ ⇒ Biri sağlanmıyorsa kanal bozulmalı (with distortion)

$\theta_n(f) = -2\pi f t_d$

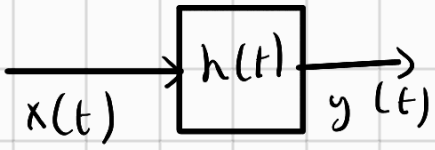
kanal bozulmalı ise $x(t)$ zamanda yayılır (time dispersion)

Örnek: (Lathi)

$$H(f) = \begin{cases} (1 + k\cos(2\pi f t))\, e^{-j2\pi f t_d}, & |f| < B \\ 0, & \text{Else} \end{cases}$$



A LPF transfer function is given in the figure. A pulse $x(t)$ band-limited to B Hz is applied at the input of this filter. Find the output $y(t)$



$$Y(f) = X(f)H(f) = X(f)\left[1 + k\cos(2\pi f T)\right] e^{-j2\pi f t_d}$$

$\cos(2\pi f T) \rightarrow \frac{1}{2}\left[e^{j2\pi f T} + e^{-j2\pi f T}\right]$

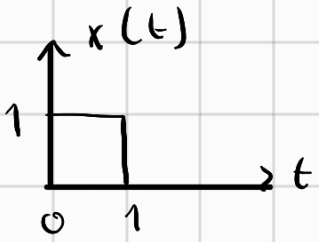
$$Y(f) = X(f)\, e^{-j2\pi f t_d} + k\, X(f)\cos(2\pi f T)\, e^{-j2\pi f t_d}$$

$\updownarrow F^{-1}$

$$y(t) = x(t-t_d) + \frac{k}{2}\left[x(t+T-t_{d1}) + x(t-T-t_d)\right]$$

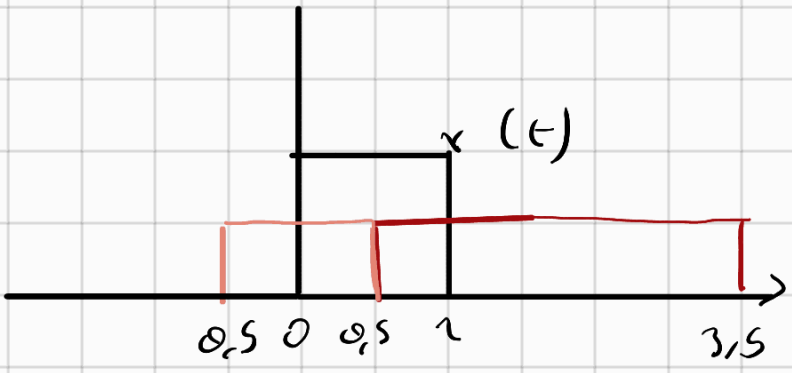
$$x(t) + \frac{k}{2}\left[x(t+T) + x(t-T)\right] \xrightarrow{\text{td seçilse}}$$

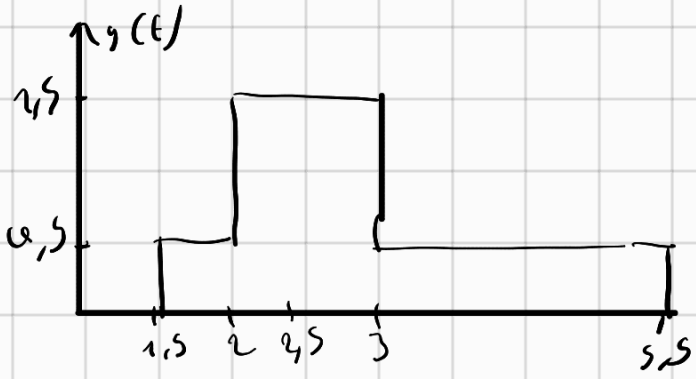
e.g



$k=1,\ T=0,5,\ t_{d1}=2$

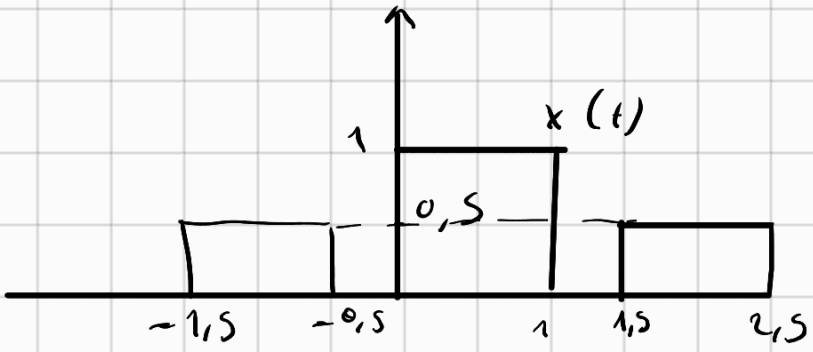
Burada yanlışlık var //



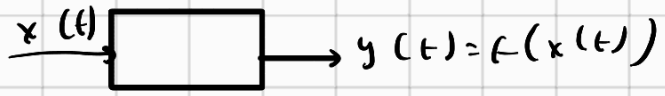


eg

$k=1,\ T=1,5,\ t_{d1}=2$



- Linear olmayan bozulma
(Non linear distortion)



Belleksiz lineer
olmayan kanal
(memoriless non linear channel)

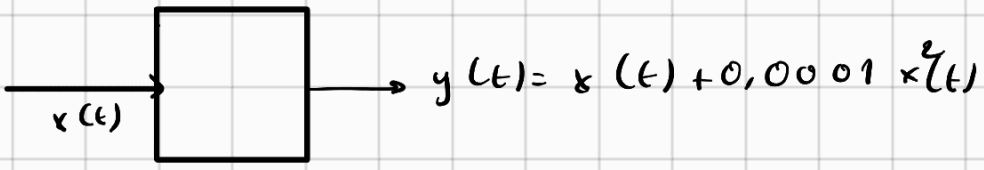
Maclaurin serisine açılırsa

$$y(t) = f(x(t)) = a_0 + a_1 x(t) + a_2 x^2(t) + a_3 x^3(t) + \cdots + a_k x^k(t)$$

$$BG\{x(t)\} = B \qquad BG\{x^2(t)\} = k \cdot B \text{ Hz}$$

⇒ Frekansta yayılma
(spectral dispersion)

Örnek: (Lathi)



$x(t) = 2000\,\text{sinc}(2000t)$

$y(t) = ?$ $\quad Y(f) = ?$ $\quad BW\{y(t)\} = ?$

Can the signal x (t) be recovered without distortion from the output y (t)?

$$A\,2\,\text{sinc}(t2) \xleftrightarrow{F} A\,\Pi\left(\frac{f}{2}\right)$$

$$\frac{2000}{2}\,\text{sinc}\left(\frac{2000t}{2}\right) \xleftrightarrow{F} X(f) = \Pi\left(\frac{f}{2000}\right)$$

$$y(t) = x(t) + 0,0001\,x^2(t)$$
$$= x(t) + 400\,\text{sinc}^2(2000t)$$

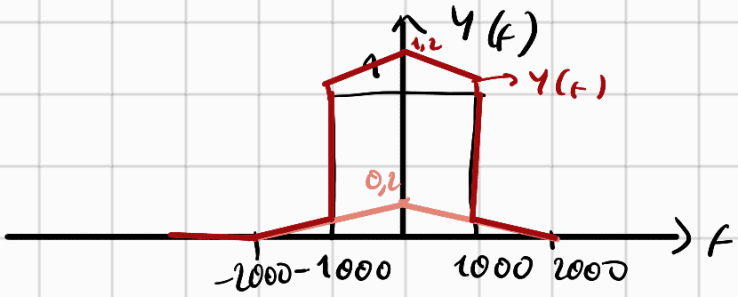
$$A\,2\,\text{sinc}^2(t2) \xleftrightarrow{F} A\,\Lambda\left(\frac{f}{2}\right)$$

$\sin^2(2000t) \xrightarrow{\mathcal{F}} 0{,}2\,\Lambda\left(\frac{f}{2000}\right)$

$A = \frac{2}{\frac{2000}{1000}} \cdot 400$

$y(t) = 2000\,\mathrm{sinc}(2000t) + 400\,\mathrm{sinc}^2(2000t)$

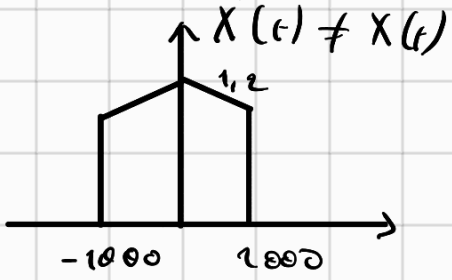
$Y(f) = \Pi\left(\frac{f}{2000}\right) + 0{,}2\,\Lambda\left(\frac{f}{2000}\right)$ ✓



$BW\{y(t)\} = 2000\,Hz$



$B_0 = BG\{x(t)\} = B\,Hz$



Bozulma

- Lineer bozulma → zamanda yayılma → TDM'de karışım (interference)
- Lineer olmayan bozulma → Frekansta yayılma → FDM'de karışım
- Çok yollu bozulma (multipath distortion)

Modülasyon (Modulation)

Temel band işareti (Base band signal) → Baseband comm. Temel bant haberleşmesi (dijitalde yapılır)

Voice(ses): 0-4kHz
Video(görüntü): 0-5/6 MHz (siyah-beyaz)

} Baseband analog signals = message signals

Modülasyon → Carrier communication
(Taşıyıcı haberleşme)

Alçak geçiren veya bant tabanlı işaretin bg. hale geçirmektir.

$BG\{m(t)\} = B$, $M(f) = 0$, $|f| > B$ ⇒ Temel band işareti (Low pass signal)

Taşıyıcı işaret (Carrier signal)

$$c(t) = A_c \cos(2\pi f_c t + \theta_c)$$

- A_c ↙ Genlik (Amplitude)
- f_c ↓ Frekans (Freq)
- θ_c ↘ Faz (phase)

Frekans + Faz → Açı modülasyonu

Modülasyonun Sebepleri:

1) Frekans bölmeli çoklama (FDÇ) yapmak için
(Frequency Division multiplexing (FDM))
BTK haberleşme bandını söylüyor

2) Verici alıcı yapısını basitleştirmek için.
Dalga boyu küçülür
ℓ = Anten boyu $\propto \frac{\lambda}{4}$ → Dalga boyu (wavelength)
length of antenna

Ses BG ≅ 3kHz $\quad c = f \cdot \lambda \Rightarrow \lambda = \frac{c}{f}$

$$= \frac{3 \cdot 10^8 \text{ m/sn}}{3 \cdot 10^3 \text{ Hz}}$$
↳ 1/sn

$= 10^5 m = 100 km \rightarrow \ell \cong \frac{\lambda}{4} = 25 km$

Genlik modülasyonu (yarıçık) : $f = 1 MHz$

$$\lambda = \frac{3 \cdot 10^8 m/sn}{10^6 Hz} = 300 m \rightarrow \ell \cong \frac{\lambda}{4} = 75 m$$

3) Mesaj işaretinin band genişliği arttırılarak gürültü ve karışımı bastırma özelliğini arttırmak için (noise and interference immunity)
→ Açı modülasyonu